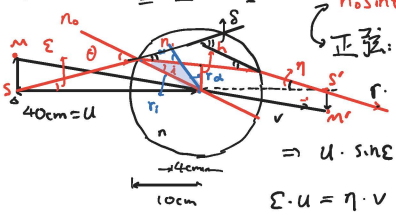
7.1. $r\cdot\sin\theta\cdot n\cdot\sin\alpha = C$

$n_0\sin\theta = n_1\sin i$

正弦: $r_1\sin i = r_2\sin\alpha$

$r\cdot n\cdot\sin\theta = \text{Const}$



$\Rightarrow u\cdot\sin\Sigma\cdot 1 = h\cdot\sin 90^\circ\cdot n_1$

$\Sigma\cdot u = \eta\cdot v = n_1 h.$

$\delta = \Sigma+\eta-2(\theta-i)$

$= 2(n-1)\cdot\theta = 2\cdot(n-1)\cdot\frac{\Sigma u}{r_1}$

$r_1=10cm$

$r_2=4cm$

(1) $\Sigma = \frac{n_1 h}{u}$, $\eta = \frac{n_1 h}{v}$

$\Rightarrow \frac{n_1 h}{u}+\frac{n_1 h}{v} = 2\cdot(n-1)\cdot\frac{h}{r_1}$

$\frac{1}{u}+\frac{1}{v} = \frac{2\cdot(n-1)}{n\cdot r_1} = \frac{1}{f}$.

$\frac{2\cdot 1.33}{1.33\cdot 10}$

小球轨迹~光轨迹.

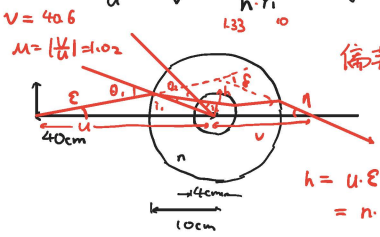
$v\to n\cdot c_0$

$n(r)\to V(r) = -\frac{1}{2}mc_0^2 n^2(r)$

初速角 $\to n(r_0)c_0\sin\theta$

$v = 40.6$

$M = |\frac{u}{v}| = 1.02$



偏转角 δ.

$\delta = (\theta_1 - i_1 + \theta_2 - i_2)\cdot 2$

$h = u\cdot\Sigma = \theta_1 r_1 = n\cdot i_1\cdot r_1$

$= n\cdot\theta_2\cdot r_2 = i_2\cdot 1\cdot r_2$

$= \eta\cdot v$

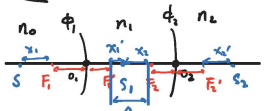
$\delta = \Sigma+\eta$

$\Rightarrow \frac{h}{u}+\frac{h}{v} = 2\left(\frac{h}{r_1}-\frac{h}{nr_1}+\frac{h}{nr_2}-\frac{h}{r_2}\right)$

$\frac{1}{u}+\frac{1}{v} = 2\cdot\frac{n-1}{n r_1} - 2\frac{n-1}{n r_2} = \frac{1}{f}$

$u=-10.01$ $\quad -0.0744$ $\quad |\frac{v}{u}| = 0.25$

透镜组.



$\frac{n_0}{f_1+x_1}+\frac{n_1}{f_1'+x_1'} = \phi_1$

$f_1 = \frac{n_0}{\phi_1}$, $f_1' = \frac{n_1}{\phi_1}$

$\frac{n_0}{\frac{n_0}{\phi_1}+x_1}+\frac{n_1}{\frac{n_1}{\phi_1}+x_1'} = \phi_1$

$n_0 x_1' + \frac{n_0 n_1}{\phi_1} + n_1 x_1 + \frac{n_1 n_0}{\phi_1} = n_1 x_1' + x_1 x_1'\phi_1 + n_0 x_1 + x_1 x_1'\phi_1$

$\Rightarrow x_1\cdot x_1' = \frac{n_0}{\phi_1}\cdot\frac{n_1}{\phi_1} = f_1 f_1'$

$x_1\cdot x_1' = f_1\cdot f_1'$ $\quad x_1'+x_2 = \Delta = d - f_1' - f_2$

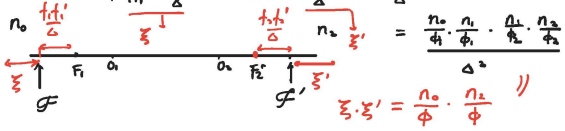
$x_2\cdot x_2' = f_2\cdot f_2'$

$\Rightarrow \frac{f_1 f_1'}{x_1}+\frac{f_2 f_2'}{x_2'} = \Delta$

$\Rightarrow x_2' f_1 f_1' + x_1 f_2 f_2' = \Delta\cdot x_1\cdot x_2'$

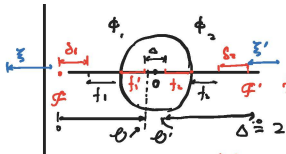
$\Rightarrow \Delta\cdot\left(x_1-\frac{f_1 f_1'}{\Delta}\right)\left(x_2'-\frac{f_2 f_2'}{\Delta}\right) = \frac{f_1 f_1'}{\Delta}\cdot f_2 f_2'$

$\left(x_1-\frac{f_1 f_1'}{\Delta}\right)\cdot\left(x_2'-\frac{f_2 f_2'}{\Delta}\right) = \frac{f_1 f_1'}{\Delta}\cdot\frac{f_2 f_2'}{\Delta}$



$\xi\cdot\xi' = \frac{n_0}{\phi}\cdot\frac{n_2}{\phi}$ //

$\Rightarrow \phi^2 = \phi_1^2\phi_2^2(\Delta/n_1)^2$ $\quad \frac{n_0 n_2}{\phi^2} = \frac{n_0}{\phi_1^2}\cdot\frac{n_2}{\phi_2^2}\cdot\frac{1}{(\Delta/n_1)^2}$



$\phi_1 = \phi_2 = \frac{n-1}{r}$

$f_1' = f_2 = \frac{n}{\phi_1} = \frac{nr}{n-1}$

$\Delta = 2r - \frac{2nr}{n-1} = \frac{-2}{n-1}\cdot r$

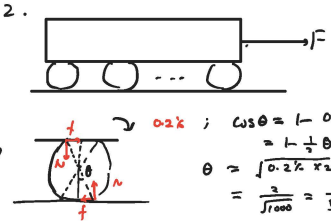
$\delta_1 = \frac{f_1 f_1'}{\Delta} = \frac{1\cdot r}{n-1}\cdot\frac{nr}{n-1}\cdot\frac{n-1}{-2} = -\frac{nr}{2(n-1)}$

$\xi\cdot\xi' = \frac{f_1 f_1'}{\Delta}\cdot\frac{f_2 f_2'}{\Delta}$

$= \left(\frac{-n}{2(n-1)}r\right)^2 = f^2$

新光心 $e_0 = \delta_1 + f_1 + r - f = 0$

2.



$\cos\theta = 1-0.2k$

$= 1-\frac{1}{2}\theta^2$

$\theta = \sqrt{0.2k\times 2}$

$= \frac{2}{\sqrt{1000}} = \frac{1}{5\pi_0} \ll 1$

$\Rightarrow \Sigma M = 0$

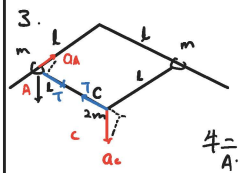
$f = N\cdot\theta$

$f_{max} = \Sigma f = \Sigma N\cdot\theta = mg\cdot\theta$

$= 6.2\times 10^4 N$

$\mu = \frac{f_{max}}{mg} = \theta = 0.06.$

3.



(1) $v=0$

a_A. a_c 沿绳分量相同.

$a_A\cdot\frac{1}{2} = a_c\cdot\frac{1}{2}$.

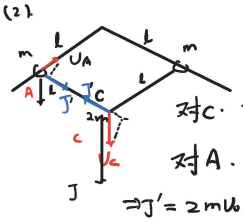
牛二: A. $m\cdot a_A = T\frac{1}{2} - mg\cdot\frac{1}{2}$. ①

C. $2m\cdot a_c = 2mg - T\cdot\frac{1}{2}\cdot 2$. ②

②+2①. $4m a_c = mg$.

$a_c = \frac{g}{4} = a_A$. $T = \frac{3}{2}mg$.

(2)



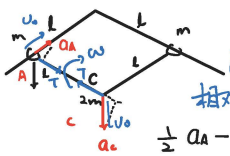
$U_A\cdot\frac{1}{2} = U_c\cdot\frac{1}{2} = \frac{1}{2}U_0$

对C. $J - J'\frac{1}{2}\cdot 2 = 2m\cdot U_0$

对A. $J'\cdot\frac{1}{2} = m\cdot U_0$

$\Rightarrow J' = 2mU_0$.

$J = 4mU_0$.



$\omega\cdot l = U_0\frac{\sqrt{3}}{2} + U_0\frac{\sqrt{3}}{2} = \sqrt{3}U_0$

$\vec{a}_B\cdot\hat{n} - \vec{a}_c\cdot\hat{n} = \omega^2 l$.

相对是圆周运动.

$\frac{1}{2}a_A - \frac{1}{2}a_c = \frac{3U_0^2}{l}$ $\quad a_A = a_c + \frac{6U_0^2}{l}$

牛二. 对A. $\frac{T}{2} - \frac{mg}{2} = m a_A$ ①

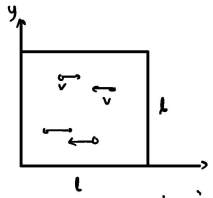
对C. $2mg - T\cdot\frac{1}{2}\cdot 2 = 2m a_c$ ②

2①-②. $2T - 3mg = 2m\cdot\frac{6U_0^2}{l}$.

$T = \frac{3}{2}mg + m\frac{6U_0^2}{l}$

4. n. m_0. v.

(1) y方向以u运动. S_y



S_y 中: $u_x' = \frac{\sqrt{1-u^2/c^2}\,v}{1-0}$

$u_y' = u$

一次来回时: $t = \frac{2l}{u_x'} = \frac{2l}{\sqrt{1-u^2/c^2}\cdot v}$

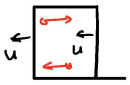
一次交换动量.

$p_1 = 2m\cdot\frac{u_x'}{\sqrt{1-\frac{u_x'^2+u_y'^2}{c^2}}} = 2\frac{m\cdot v}{\sqrt{1-\frac{u^2}{c^2}}}$

$p_{压} = \frac{\text{单位时间交换动量}}{\sqrt{1-u^2/c^2}\cdot l^2} = \frac{n l^2\cdot\frac{2mv}{\sqrt{1-u^2/c^2}}}{\sqrt{1-u^2/c^2}\cdot l^2}$

$p_{压y} = n\cdot\frac{mv^2}{\sqrt{1-v^2/c^2}}$ 变

(2) $u_{xR} = \frac{v-u}{1-\frac{vu}{c^2}}$. $u_{xL} = \frac{v+u}{1+\frac{vu}{c^2}}$.



$t_R = \frac{l/\gamma_u}{u_{xR}+u} = \frac{\sqrt{1-u^2/c^2}\,l}{\frac{v-u}{1-\frac{vu}{c^2}}+u}$

$= \frac{\sqrt{1-u^2/c^2}\cdot l\cdot(1-\frac{vu}{c^2})}{v-u+u-\frac{u^2}{c^2}v} = \frac{l(1-\frac{vu}{c^2})}{\sqrt{1-u^2/c^2}\cdot v}$

$t_L = \frac{l/\gamma_u}{u_{xL}-u} = \frac{l(1+\frac{vu}{c^2})}{\sqrt{1-u^2/c^2}\,v}$

$t = t_R + t_L = \frac{2l}{\sqrt{1-u^2/c^2}\,v}$

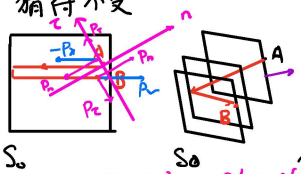
一次交换 $p_1 = \frac{mu_R}{\sqrt{1-\frac{u_R^2}{c^2}}} + \frac{mu_L}{\sqrt{1-\frac{u_L^2}{c^2}}}$

$= \frac{m\cdot\frac{v-u}{1-\frac{vu}{c^2}}}{\sqrt{1-(\frac{v-u}{1-\frac{vu}{c^2}})^2}} + c.c.$

$= \frac{m\cdot(v-u)}{\sqrt{(1-\frac{u^2}{c^2})(1-\frac{u^2}{c^2})}} + c.c. = \frac{2mv}{\sqrt{1-\frac{u^2}{c^2}}\cdot\sqrt{1-\frac{u^2}{c^2}}}$

$p_{压x} = \frac{n l^2\cdot\frac{2mv}{\sqrt{1-u^2/c^2}\cdot\sqrt{1-u^2/c^2}}}{l^2\cdot\frac{2l}{\sqrt{1-u^2/c^2}\,v}} = \frac{nmv^2}{\sqrt{1-u^2/c^2}}$ 不变

(3). 猜得不变



A.B事件→钟慢

$\Rightarrow t = \frac{2l}{\sqrt{1-u^2/c^2}\,v}$

p_z 不变. $p_n' = \gamma(p_n + \beta E_c)$.

$p\sin\theta \leftarrow -p_n' = \gamma(-p_n + \beta E_c) \rightarrow p\cos\theta$

交换动量. $2p_z\cdot\hat{\tau} + 2\gamma p_n\cdot\hat{n}$

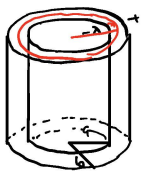
面积 $\vec{S} = S_n\hat{n} + S_z\hat{\tau} \Rightarrow \vec{S}' = \underset{S_{\theta}\sin\theta}{\frac{S_\theta}{\gamma}}\hat{\tau} + \underset{S\cos\theta}{S_n}\hat{n}$

$\Rightarrow \vec{S}' \parallel \vec{dP}'$; $p_{压} = \frac{|dP'|}{t\,|S'|} =$ 不变

S_0 中一个粒子 $\begin{bmatrix} E/c \\ \vec{p} \end{bmatrix}$ $\begin{bmatrix} E/c \\ p_x \\ p_y \\ p_z \end{bmatrix}$ 沿 x 轴换系 $\Rightarrow$

$\begin{cases} E/c = \gamma(E/c - \beta p_x) \\ p_x' = \gamma(p_x - \beta E/c) \\ p_y' = p_y \\ p_z' = p_z \end{cases}$

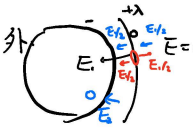
5.



$\Delta\varphi = \int_a^b E\,dx = \int_a^b \frac{\lambda}{2\pi\varepsilon_0 x}\,dx = \frac{\lambda}{2\pi\varepsilon_0}\ln b/a$ $\ln\frac{b}{a} = \ln(1+\frac{b-a}{a}) \simeq \frac{b-a}{a}$

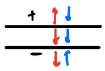
$\simeq \frac{\lambda}{2\pi\varepsilon_0}\frac{b-a}{a}$

外



$E_1 = \frac{1}{\varepsilon_0}\frac{\lambda}{2\pi b}$. $\because \sigma_1 = \frac{\lambda}{2\pi b}$.

$\frac{\Delta F}{\Delta S} = E_1\sigma_1 x$.



$\sigma_+\cdot E_- = \frac{\Delta F}{\Delta S}$. 一小团电荷·其之电荷产生场·

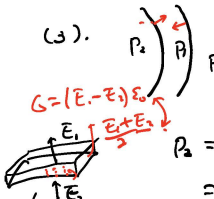
$\Rightarrow \frac{1}{2}E_1\cdot\sigma_1 = \frac{\Delta F}{\Delta S}$ ✓

$\frac{\Delta F}{\Delta S} = \frac{1}{2}\sigma\cdot\frac{\lambda^2}{\varepsilon_0^2 4\pi^2 b^2}$

内: $\frac{\Delta F}{\Delta S} = \frac{1}{2}\varepsilon_0 E_2^2 = \frac{1}{2}\varepsilon_0\left(\frac{\lambda}{\varepsilon_0 2\pi a}\right)^2$

$= \frac{\lambda^2}{2\varepsilon_0 4\pi^2 a^2}$

(3).



$P_1 = P_0 + \frac{1}{2}\varepsilon_0 E_1^2$

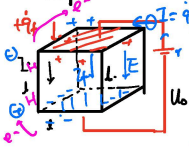
$= P_0 + \frac{1}{2}\frac{\lambda^2}{\varepsilon_0 4\pi^2 b^2}$.

$P_2 = P_1 - \frac{1}{2}\varepsilon_0 E_2^2$

$= P_0 + \frac{1}{2}\frac{\lambda^2}{\varepsilon_0 4\pi^2 b^2}\cdot -\frac{1}{2}\frac{\lambda^2}{\varepsilon_0 4\pi^2 a^2}$

$\frac{\Delta F}{\Delta S} = \frac{1}{2}\varepsilon_0 E_1^2 - \frac{1}{2}\varepsilon_0 E_2^2$.

6.



$\vec{U}_+$. $\vec{f}_+ = -k_+\vec{U}_+$

$\vec{U}_-$. $\vec{f}_- = -k_-\vec{U}_-$ $\Sigma F = 0$

$\vec{E}$; $U_+ = \frac{1}{k_+}\cdot E\cdot 2q$.

$U_- = \frac{1}{k_-}E\cdot q$.

$j_+ = n_0\cdot 2q\cdot\frac{E}{k_+}\cdot 2q$.

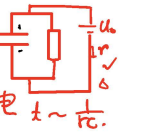
$j_- = 2n_0\cdot q\cdot\frac{E}{k_-}\cdot q$

$\vec{j} = n_0\left(\frac{4q^2}{k_+} + \frac{2q^2}{k_-}\right)\cdot\vec{E}$

$I = \vec{j}\cdot\vec{S} = l^2\cdot n_0 q^2\left(\frac{4}{k_+}+\frac{2}{k_-}\right)\cdot\frac{U_0}{L}$ $t\to 0^+$

$\sim R = \frac{1}{L\,n_0 q^2\left(\frac{4}{k_+}+\frac{2}{k_-}\right)}$;

等效电容: $C = \frac{\varepsilon_0 l^2}{L} = \varepsilon_0 l$.



瞬间充电 $t \sim \frac{1}{Cr}$.

极高频 来不及复合

$m\vec{a} = \vec{E} + \vec{f}$

$m\vec{a} = \vec{E}_0 - k\vec{v}$ 本征频

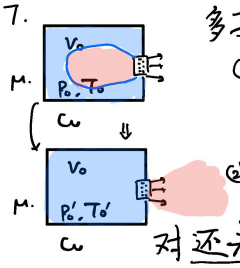
一维振动



$m\ddot{x} = E\cos\omega t\cdot 2q - \frac{2nq}{\varepsilon_0}x - 2q - kx$

$m\cdot j = -E\cos\omega t\cdot q + \frac{2nq}{\varepsilon_0}y\cdot q - ky$

7. 多孔塞



① 两边有 Δp 时. 从慢速流动气体. → 几乎准静态.

② 几乎绝热

对还未流出气体写绝热方程

热 — 状态 — 受力 — 热流

每一个时刻压强均匀 ↓ T均匀 → 热流=0.

$$P_0\cdot\left(\frac{V_0}{2}\right)^\gamma = P_0'\cdot V_0^\gamma$$

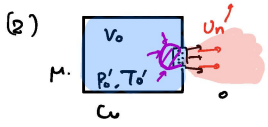
$$P_0' = P_0\cdot 2^{-\frac{5}{3}} \qquad \gamma = \frac{C_p}{C_v} = \frac{5}{3}$$

$$\begin{cases} pV^\gamma = C \\ pV = nRT \end{cases} \rightarrow p^{\gamma-1} = C\cdot T^\gamma$$

$$\left(\frac{T_0'}{T_0}\right) = \left(\frac{P_0'}{P_0}\right)^{\frac{\gamma-1}{\gamma}} = \left(2^{-\frac{5}{3}}\right)^{\frac{2}{5}} = 2^{-\frac{2}{3}}$$

$$T_0' = 2^{-\frac{2}{3}}T_0$$ 相对静止

(2)



剩余摩尔数 $n \to n+dn$, $dn<0$

$$n_0 = \frac{P_0V_0}{RT_0}$$

$$\left(\frac{n}{n_0}V_0\right)^\gamma P_0 = P_n\cdot V_0^\gamma \Rightarrow P_n = P_0\left(\frac{n}{n_0}\right)^{\frac{5}{3}}$$

$$\left(\frac{T_n}{T_0}\right) = \left(\frac{n}{n_0}\right)^{\frac{2}{3}}$$

$$\frac{1}{2}(-dn)\cdot\mu\cdot u_n^2 = dV\cdot P_n = (-dn)\cdot RT_n$$

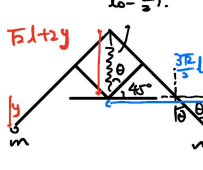
$$\Rightarrow u_n = \sqrt{\frac{2RT_n}{\mu}}$$

$$\Rightarrow P_{动} = \int_{n_0}^{0} -dn\cdot\mu\cdot u_n = \int_0^{n_0}\mu\cdot\sqrt{\frac{2R}{\mu}\cdot T_0\left(\frac{n}{n_0}\right)^{\frac{1}{3}}}\cdot dn$$

$$\int_0^1 x^{1/3}dx = \frac{3}{4}x^{\frac{4}{3}}$$

$$= \sqrt{2\mu RT_0}\cdot n_0\cdot\frac{3}{4} = \sqrt{2\mu RT_0}\,\frac{3}{4}\cdot\frac{P_0V_0}{RT_0}$$

8.



3:3:1:1 $\frac{\sqrt{2}}{2}l$ 弹簧

单一自由度 θ.

$$E_p = -2lmg\cos\theta\cdot + \frac{1}{2}k\cdot(2l\cos\theta - \sqrt{2}l)^2$$

$$\left.\frac{dE_p}{d\cos\theta}\right|_{\theta=45^\circ} = 0 \qquad \cos\theta = \frac{\sqrt{2}}{2}$$

$$\Rightarrow -2lmg + k\cdot(2l\cos\theta - \sqrt{2}l)\cdot 2l\cdot$$

$$2l\cdot k\cdot\sqrt{2}l = 2lmg \Rightarrow k = \frac{\sqrt{2}mg}{l}$$

动: $\theta\sim\theta+\varepsilon$. 二阶 $l\cos\theta = \frac{\sqrt{2}}{2}l + y$.

$$E_p = -2mg\left(\frac{\sqrt{2}}{2}l+y\right) + \frac{1}{2}k\cdot\left(\sqrt{2}l+2y-\frac{\sqrt{2}}{2}l\right)^2$$

$$E_p = -2mg\left(\frac{\sqrt{2}}{2}l+y\right) + \frac{1}{2}k\cdot\left(\sqrt{2}l+2y+\frac{\sqrt{2}}{2}l\right)^2$$

$$= \square + \frac{1}{2}\cdot 2ky^2 \quad \Delta$$

$$E_k = \frac{1}{2}m(\dot x^2+\dot y^2)\times 2\cdot = \frac{1}{2}\cdot 2m\cdot 10\dot y^2 \quad \Delta$$

$$\frac{\sqrt{2}}{2}l + x = l\sin\theta \quad ; \quad \frac{\sqrt{2}}{2}l + y = l\cos\theta$$

$$\Rightarrow \left(\frac{\sqrt{2}}{2}l+y\right)^2 + \left(\frac{x}{3}+\frac{\sqrt{2}}{2}l\right)^2 = l^2$$

$$y^2 + \sqrt{2}ly + \frac{x^2}{9} + 2\cdot\frac{x}{3}\cdot\frac{\sqrt{2}}{2}l = 0$$

$$2y\dot y + \sqrt{2}l\dot y + \frac{2x}{9}\dot x + \frac{\dot x}{3}\sqrt{2}l = 0$$

$$\Rightarrow \dot y = -\frac{\dot x}{3} \qquad \dot x = -3\dot y$$

$$\begin{cases} E_p = \frac{1}{2}\cdot 2k\,y^2 \\ E_k = \frac{1}{2}\cdot 20\,m\dot y^2 \end{cases}$$

$$\Rightarrow \omega^2 = \frac{k}{10m} = \frac{1}{10}\cdot\frac{\sqrt{2}mg}{l} = \frac{\sqrt{2}}{10}\frac{g}{l}$$

$$T = 2\pi\cdot\sqrt{\frac{10l}{\sqrt{2}g}}.$$

θ: $u_x = \frac{d}{dt}(3l\sin\theta) = 3l\cos\theta\,\dot\theta$

$u_y = \frac{d}{dt}(l\cos\theta) = -l\sin\theta\,\dot\theta$ $\quad \dot\theta = \dot\varepsilon$

$$E_k = \frac{1}{2}\cdot 2m(u_x^2+u_y^2) \qquad \theta = 45^\circ+\varepsilon.$$

$$E_p = -2mgl\cos(45^\circ+\varepsilon) \qquad O(\varepsilon^2)$$

$$+\frac{1}{2}k\cdot(2l\cos(45^\circ+\varepsilon) - \sqrt{2}l)^2$$

$$\cos(45^\circ+\varepsilon) = \cos45^\circ - \sin45^\circ\cdot\varepsilon - \frac{1}{2}\cdot\cos45^\circ\cdot\varepsilon^2$$

$$E_p = -2mgl\cdot\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\varepsilon - \frac{\sqrt{2}}{4}\varepsilon^2\right) \qquad O(\varepsilon^2)$$

$$+\frac{1}{2}kl^2\left(\frac{\sqrt{2}}{2} - \sqrt{2}\varepsilon + \frac{\sqrt{2}}{2}\varepsilon^2\right)^2 \qquad O(\varepsilon^2)$$

$$= 0 + 0\cdot\varepsilon + 2mgl\frac{\sqrt{2}}{4}\varepsilon^2 + \frac{1}{2}kl^2(2\varepsilon^2 + 2\cdot\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{2}}{2}\varepsilon^2)$$